AIRWAY CLEARANCE

BRYAN TIN

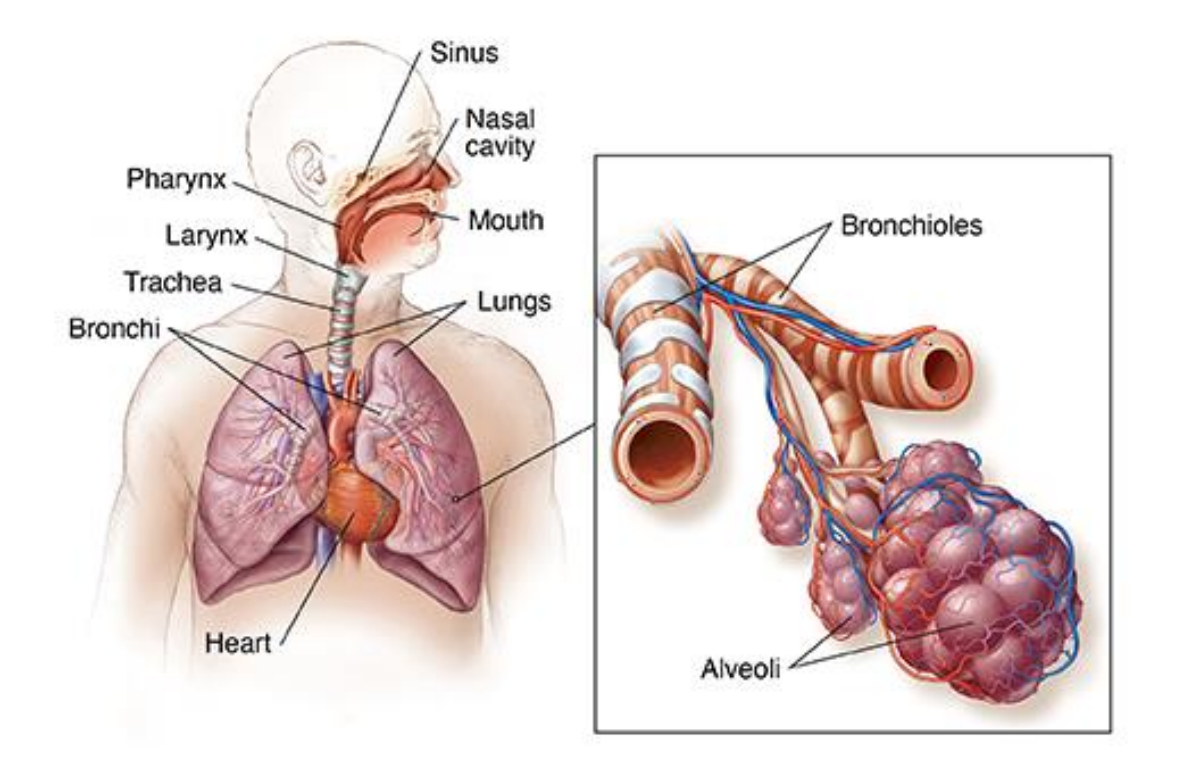
SPECIALIST RESPIRATORY PHYSIOTHERAPY

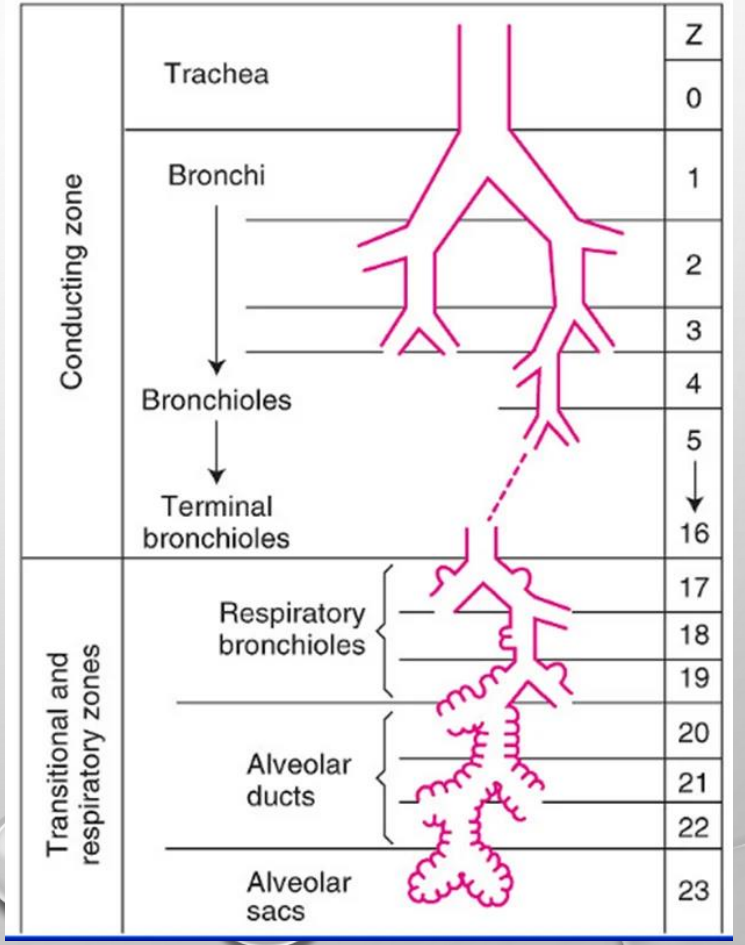
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LEARNING OUTCOMES

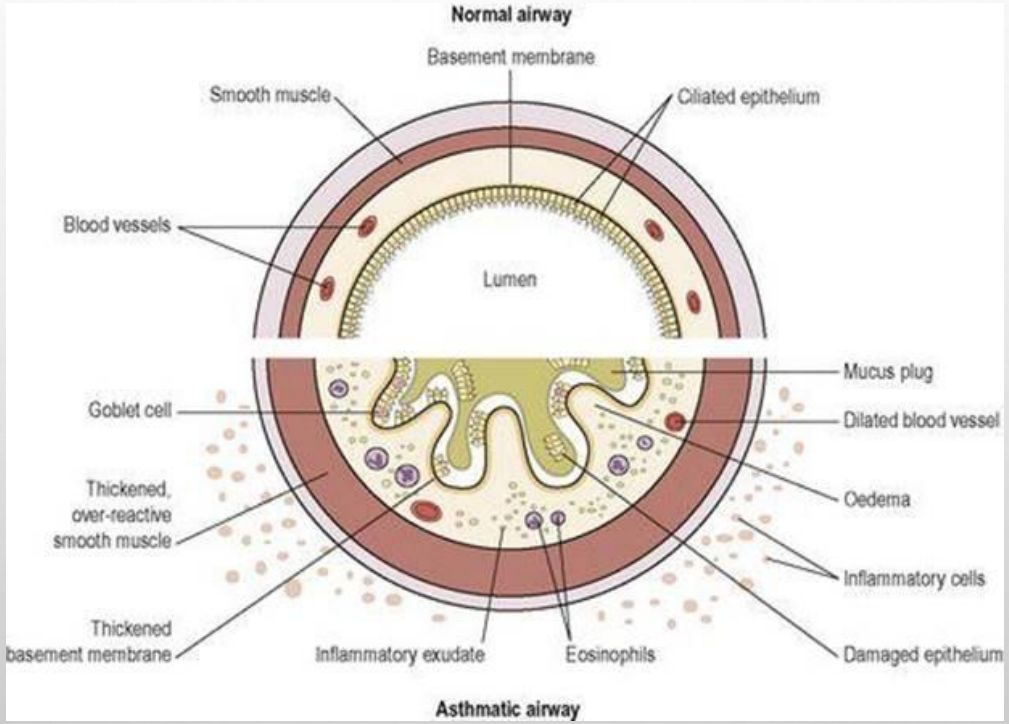
- DESCRIBE AND EXPLAIN THE PHYSIOLOGICAL PRINCIPLES BEHIND AIRWAY CLEARANCE
- UNDERSTAND PHYSIOLOGICAL PRINCIPLES OF AIRWAY CLEARANCE TREATMENTOPTIONS
- EXPLAIN THE CLINICAL REASONING BEHIND AIRWAY CLEARANCE TREATMENT SELECTION
- ABLE TO APPROPRIATELY CHOOSE TREATMENT OPTIONS FOR TARGETED PATIENTS
- EXPLAIN THE CONTRAINDICATIONS TO DIFFERENT AIRWAY CLEARANCE TECHNIQUES

PHYSIOLOGY RECAP





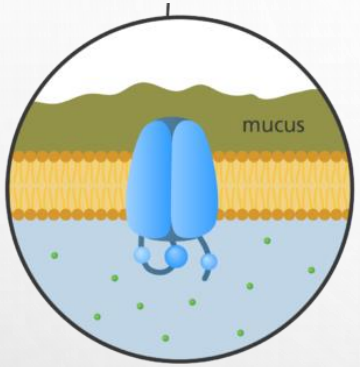
ANATOMY OF AN AIRWAY



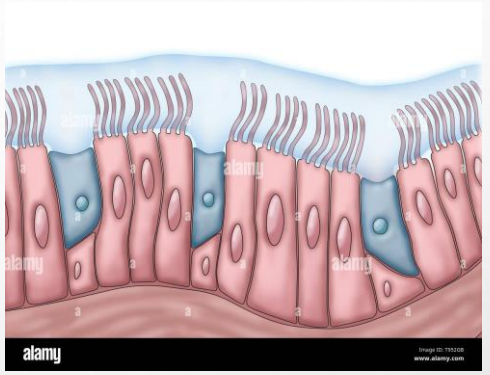
DEFINITION OF AIRWAY CLEARANCE

- THIS SYSTEM IS AN IMPORTANT LUNG DEFENCE MECHANISM CONSISTING OF AIRWAY SURFACE LIQUID COMPRISING MUCUS AND PERICILIARY LAYERS (PCLS), CILIARY EPITHELIUM AND COUGH CLEARING MECHANISMS – MCLLWAINE 2017

WHAT CAN GO WRONG



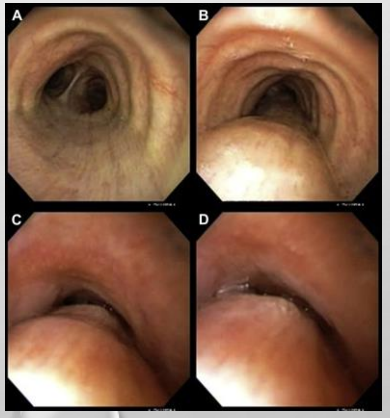
INCREASED VISCOSITY OF MUCUS



IMPAIRED CILIARY ACTIVITY



WEAKENED COUGH STRENGTH



CHANGES IN LUNG AND AIRWAY ANATOMY

PHYSIOLOGY OF EFFECTIVE AIRWAY CLEARANCE

TWO-PHASE GAS LIQUID FLOW

EXIT FLOW

EXPIRATORY COUGH FLOW BIAS

ADEQUATE COUGH STRENGTH

EQUAL PRESSURE POINT

MUCOCILIARY ESCALATOR

NORMAL MUCUS VISCOSITY

EFFECTIVE CILIA BEATING

VENTILATION

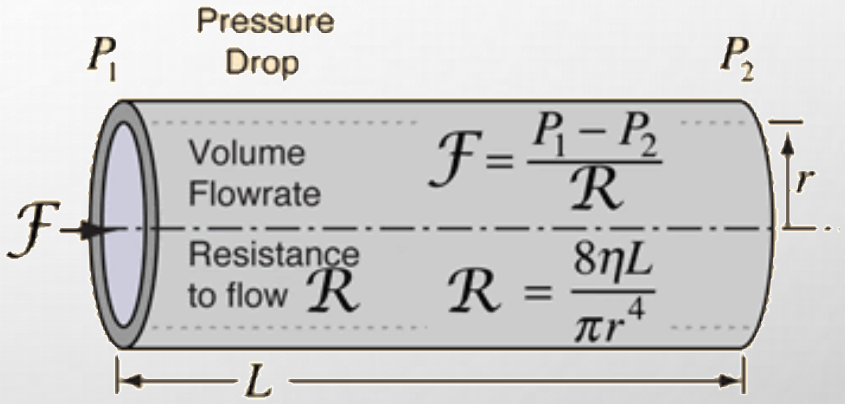
COLLATERAL VENTILATION

INTERDEPENDENCE

PEDELLUFT FLOW

AIRFLOW AND POISEUILLE'S LAW

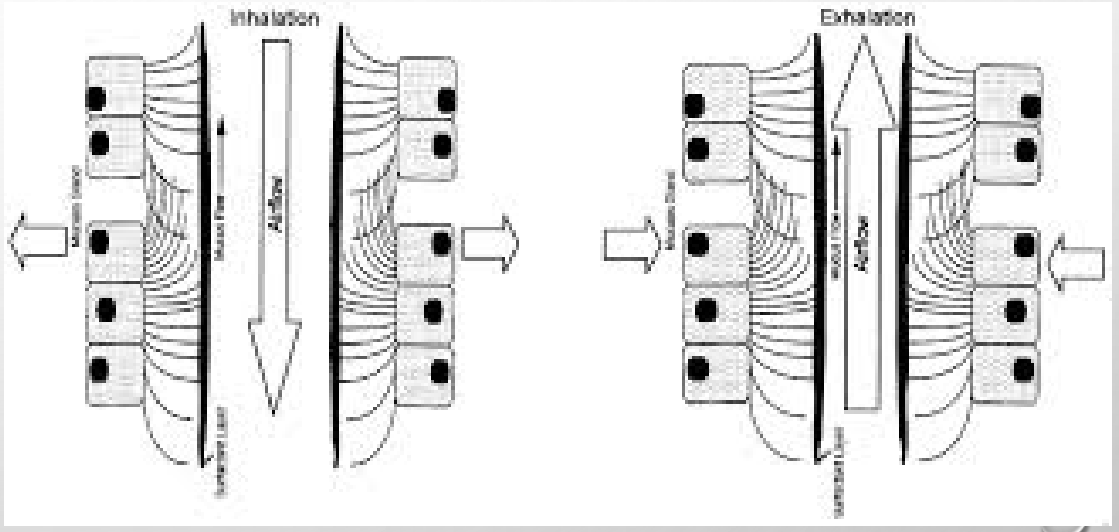
- THE FLOW OF GAS OR LIQUID THROUGH A TUBE
- VARIABLES :
  - DIAMETER OF THE TUBE
  - SPEED OF FLOW THROUGH THE TUBE
  - LENGTH OF THE TUBE
  - VISCOSITY OF LIQUID/GAS IN THE TUBE



- IN AIRWAY CLEARANCE: CHANGE OF ABOVE VARIABLES MAY IMPACT THE EFFECTIVENESS OF SPUTUM CLEARANCE

EXPIRATORY FLOW BIAS

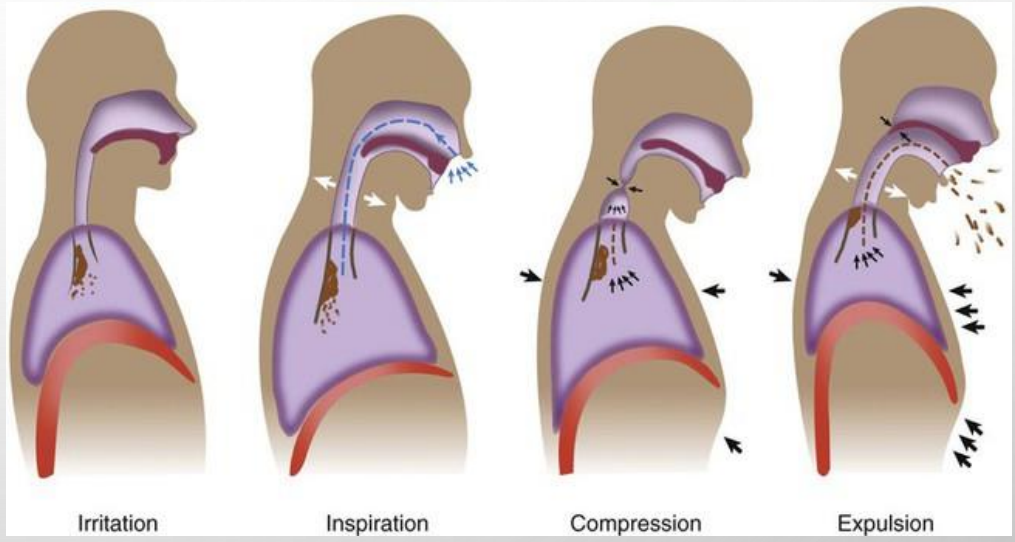
- AIRFLOW DURING EXHALATION IS GENERALLY FASTER THAN INHALATION
  - NARROWING OF AIRWAYS ON EXPIRATION
- IMPORTANCE IN MOBILIZATION OF SECRETIONS UP THE RESPIRATORY TRACT



EFFECTIVE COUGH

- COORDINATION OF COUGHING MOTION
- LARYNGEAL CONTROL
- DIAPHRAGMATIC, INTERCOSTAL AND ABDOMINAL MUSCLE CONTRACTIONS

- E.G. VOCAL CORD PALSY, NEUROMUSCULAR DISEASES



SPUTUM LOAD AND VISCOSITY

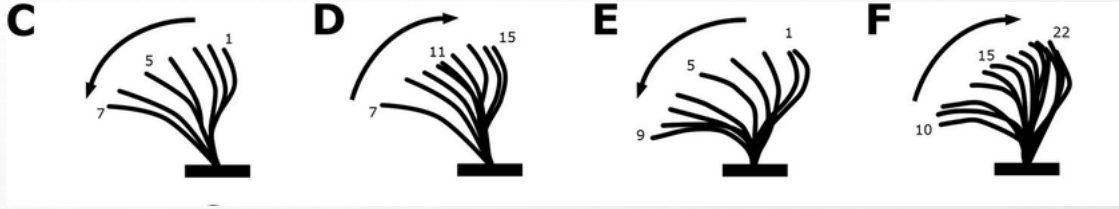
- ACT AS AIRWAY PROTECTIVE MECHANISIM
  - FILTERING AND TRAPPING PARTICLES AND PATHOGENS
  - SUPPORTING IMMUNE SYSTEM
  - MOISTURISING AND LUBRICATING AIRWAY TISSUE
- SECRETED BY GOBLET CELLS
- FACTORS THAT DETERMINES SPUTUM VISCOSITY
  - HYDRATION
  - ELECTROLYTE BALANCE
  - DISULPHIDE BONDING IN MUCIN GLYCOPROTEIN
  - LIPIDS
  - MICROBIOME
- E.G BACTERIAL COLONIZATION, CF, PLASTIC BRONCHITIS

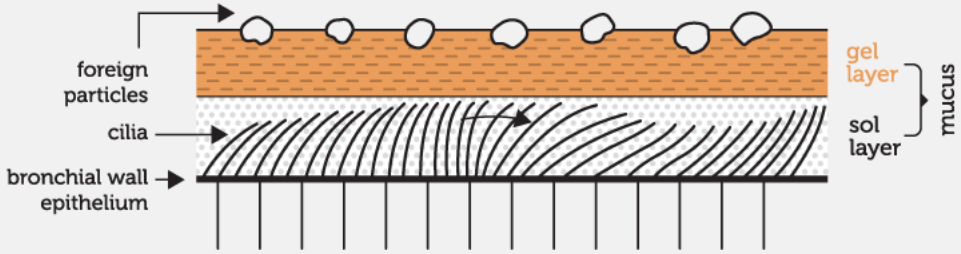




CILIA BEATING

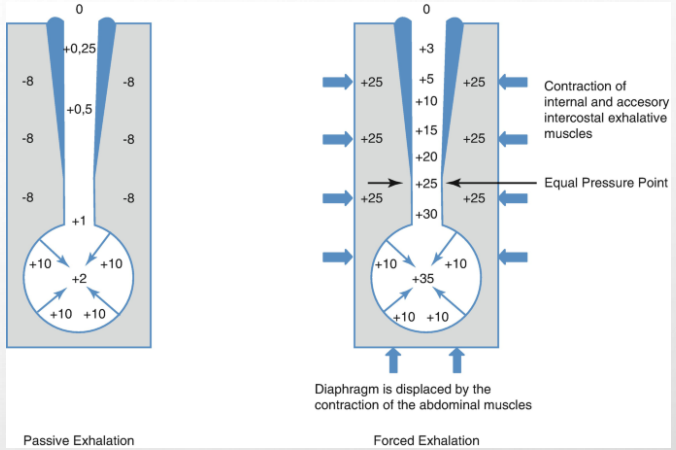
- TO FACILITATE MOBILIZATION OF MUCUS UP THE RESPIRATORY TRACT
- UNIDIRECTIONAL
- 11-18HZ
- E.G PCD

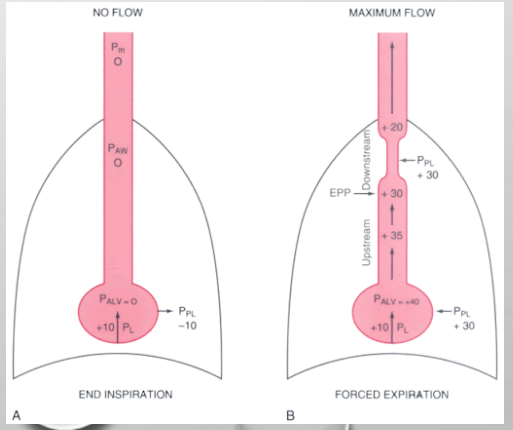




EQUAL PRESSURE POINTS

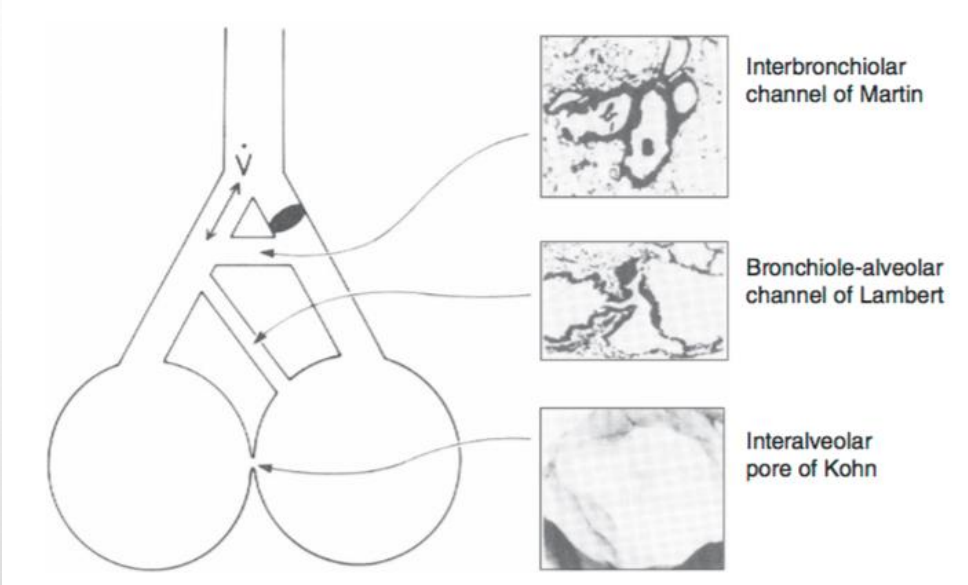
- A PINCH POINT WHERE AIRWAY PRESSURE MATCHES PLEURAL PRESSURE
- BEYOND THIS POINT A POINT OF COLLAPSE HAPPENS
- THIS HAPPENS IN CARTILAGES SUPPORTED AIRWAY IN NORMAL HEALTH INDIVIDUALS
- CAN BE AUGMENTED AND UTILIZE TO SUPPORT AIRWAY CLEARANCE





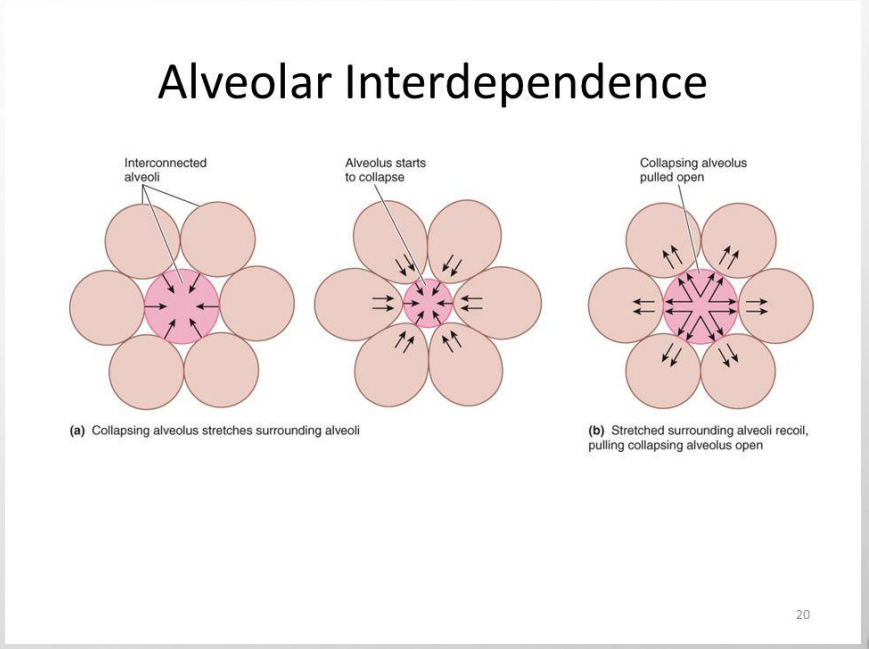
COLLATERAL VENTILATION

- STANDARD ROUTE OF AIRFLOW INTO THE ALVEOLI : TERMINAL BRONCHIOLES
- COLLATERAL VENTILATION CHANNELS:
  - CHANNELS OF LAMBERT (BRONCHIOLE TO ALVEOLI)
  - CHANNELS OF MARTIN (BRONCHIOLE TO BRONCHIOLE)
  - PORES OF KOHN (ALVEOLI TO ALVEOLI)
- ALLOW AIRFLOW PAST A BLOCKED BRONCHIOLE TO AID CLEARANCE
- NORMALLY NOT UTILISED IN HEALTHY INDIVIDUALS
- COMPLETELY DEVELOPS AT AROUND AGE OF 4



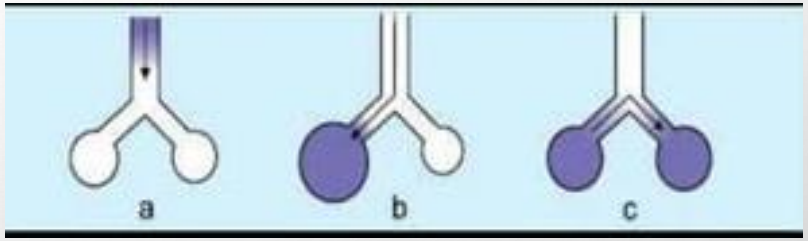
INTERDEPENDENCE

- ALVEOLI ARE CONNECTED TO ONE ANOTHER
- IF ONE ALVEOLI STARTS TO COLLAPSE THE OTHERS WILL STRETCH AND HELP TO KEEP THEM OPEN - ENHANCED TRACTION FORCE WHICH TRANSMITS TO ADJACENT ALVEOLI
- COUNTERACTING EARLY COLLAPSE OF AIRWAYS
- THIS RELIES ON THE ELASTICITY OF THE ALVEOLI AND THE LUNGS
- E.G ATELECTASIS



PENDELLUFT FLOW

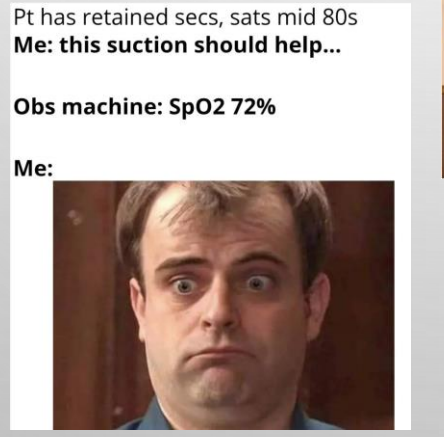
- A CONCEPT REFERS TO THE MOVEMENT OF AIR BETWEEN AREAS OF LUNG DUE TO UNEVEN VENTILATION
- AIR MOVES FROM FASTER FLOW TO SLOWER FLOW AREAS
- CREATION OF EXTRA SHEARING FORCE
- HOMOGENISE VENTILATION



NOW WE UNDERSTAND THE PHYSIOLOGY…

- LETS START TO THINK ABOUT THE WAY THAT PHYSIOTHERAPISTS USE THESE PHYSIOLOGICAL CONCEPTS TO AID AIRWAY CLEARANCE…..











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LEAST INVASIVE TO MOST INVASIVE



MOBILITY AND EXERCISE

- THE MOST SIMPLE WAY OF INCREASING AIRWAY CLEARANCE
- PATIENTS ARE IN A GOOD POSITION TO IMPROVE LUNG VOLUMES (UPRIGHT)
- INCREASES TIDAL VOLUMES NATURALLY WITH EXERCISE
- INCREASED MINUTE VENTILATION
- WORK OF BREATHING INCREASES EXIT FLOW -> SHEAR FORCE
- GET THEM SLIGHTLY BREATHLESS!
- SHOULD BE USED PRIMARY AS PREVENTATIVE NOT A RESCUE METHOD



MOBILITY AND EXERCISE

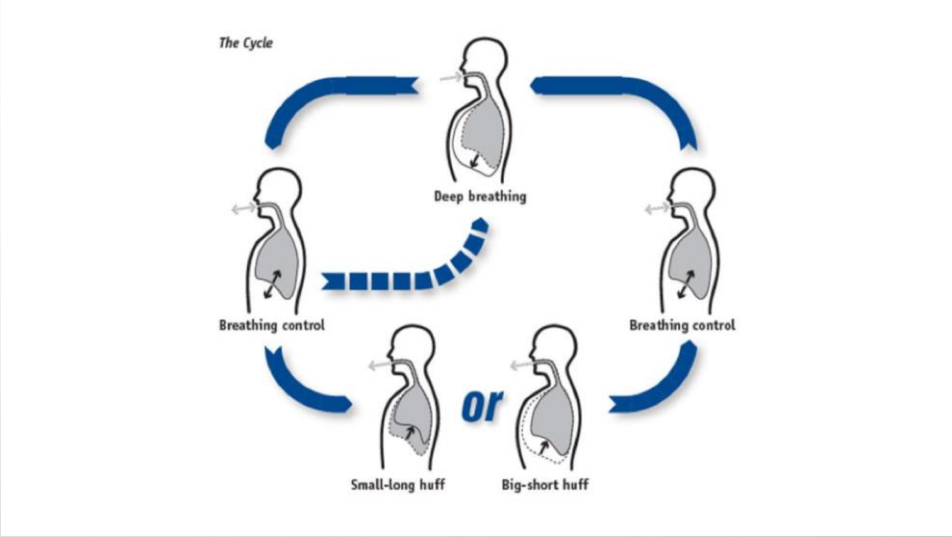
- INDICATIONS:
  - RETAINED SECRETIONS
  - ATELECTASIS

- CONTRAINDICATIONS:
  - LOW HB
  - UNSTABLE HAEMODYNAMICS

- CONSIDERATIONS :
  - OXYGEN REQUIREMENTS AND DELIVERY
  - ATTACHMENTS
  - FATIGUE
  - LEVEL OF MOBILITY AND SUPPORT NEEDED

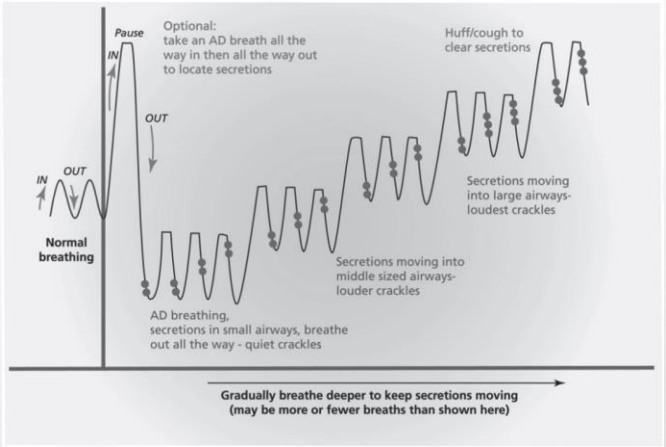
ACTIVE CYCLE OF BREATHING (ACBT)

- UTILISATION OF COLLATERAL VENTILATION, INCREASED EXPIRATORY FLOW AND POTENTIALLY EQUAL PRESSURE POINTS IF NEEDED
- 4 PARTS :
  - BREATHING CONTROL
  - THORACIC EXPANSION EXERCISES
  - FORCED EXPIRATION TECHNIQUE (HUFF – CAN BE GRADED)
  - COUGH



AUTOGENIC DRAINAGE

- OFTEN USED IN THE CF/PCD POPULATION
- UTILISES THE EQUAL PRESSURE POINT THEORY
- BREATHS AND VERY GENTLE HUFFS ARE COMPLETED FROM LOW TO HIGH LUNG VOLUMES.
- MOVES THROUGH 3 PHASES : USTICK, COLLECT AND EVACUATE
- SIGHING NOT HUFFING
- COMPLICATED AND TRICKY TO LEARN BUT VERY EFFECTIVE



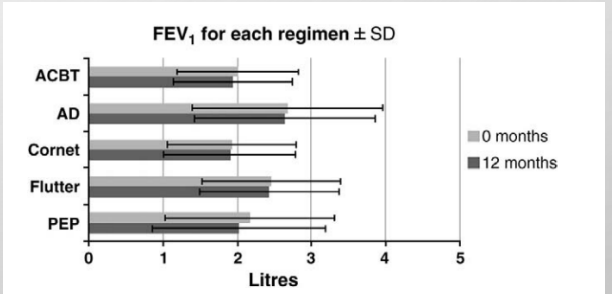


Fig. 2. FEV_1 for each regimen. Mean $FEV_1 \pm SD$: ACBT ($n=13$) at 0 months 2.01 ± 0.82, at 12 months 1.94 ± 0.80; AD ($n=13$) at 0 months 2.68 ± 1.29, at 12 months 2.64 ± 1.22; Cornet ($n=14$) at 0 months 1.93 ± 0.87, at 12 months 1.90 ± 0.89; Flutter ($n=12$) at 0 months 2.46 ± 0.94, at 12 months 2.43 ± 0.94; PEP ($n=13$) at 0 months 2.17 ± 1.14, at 12 months 2.02 ± 1.17. There were no significant differences among the regimens at 0 months or 12 months.

ACBT AND AD

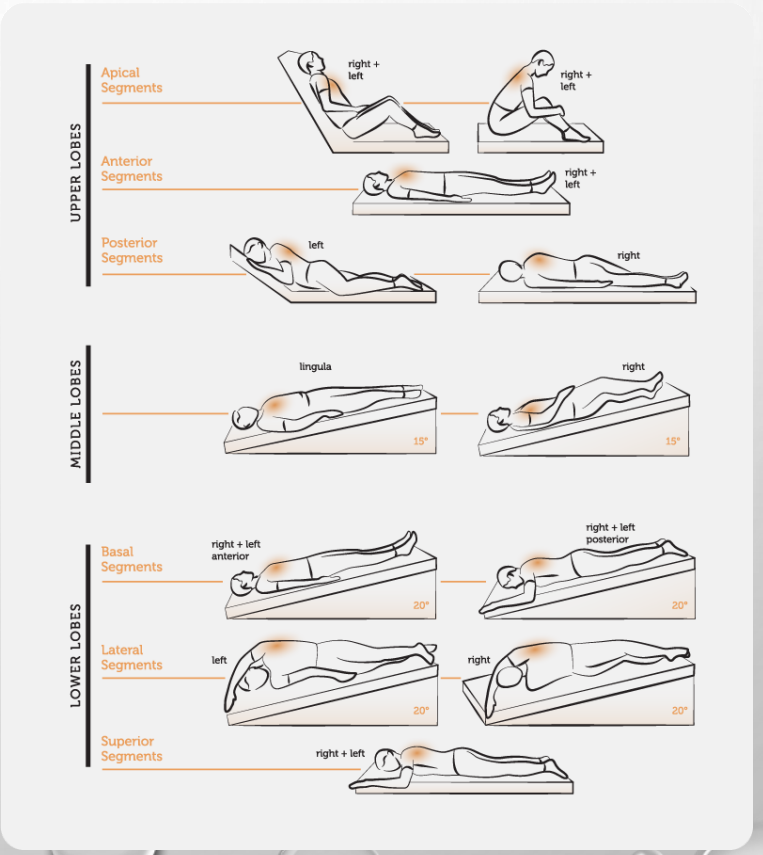
- INDICATIONS :
  - RETAINED OR THICK SECRETIONS
  - FREQUENT NEED FOR AIRWAY CLEARANCE
  - UNABLE TO COMPLETE OTHER FORMS OF AIRWAY CLEARANCE DUE TO COMPLEXITY OR MILD HAEMOPTYSIS

- CONTRAINDICATIONS:
  - FRANK HEAMOPTYSIS

- CONSIDERATIONS:
  - PAIN RELIEF
  - SUPPORT FOR SURGICAL INCISION SITES
  - LUNG SURGERY
  - SEVERE ANATOMICAL AIRWAY OBSTRUCTION

GRAVITY-ASSISTED POSITIONING

- ONE OF THE OLDEST METHOD USED
- IDEA IS TO EXPLOIT THE USE OF GRAVITY TO FACILITATE SLIDING OF MUCUS FROM PERIPHERALS TO CENTRAL AIRWAY
- NEWER STUDIES WITH RADIOACTIVE AEROSOL MARKINGS HAVE SUGGESTED GRAVITY ALONE IS UNLIKELY ABLE TO PROMOTE MUCUS DISPLACEMENT
- IT IS SUGGESTED THAT THIS INSTEAD PROMOTES VENTILATION IN TARGETED AREAS
- CONSIDER RISK FACTORS – HEAD RUSH, REFLUX, DISCOMFORT, INCREASED WOB SECONDARY TO DIAPHRAGM AND HEART DISPLACEMENT
- NOT TO USE ALONE



GAP

- INDICATIONS :
  - RETAINED SECRETIONS ON ONE SPECIFIC SIDE / AREA
  - USED WITH DIFFERENT TECHNIQUES / ADJUNCTS
  - INTUBATED OR VENTILATED PATIENTS

- CONTRAINDICATIONS
  - RESTRICTIONS ON ROLLING EG. SURGICAL,
  - NEURO-HAEMATOLOGICAL COMPLICATIONS
  - UNSTABLE HAEMODYNAMICS

- CONSIDERATIONS:
  - SURGICAL SITES
  - SURGICAL DRAINS
  - ATTACHMENTS
  - HEAD DOWN TILT : BREATHLESSNESS, CARDIAC INSTABILITY, RECENTLY EATEN , GORD, TRAUMA TO HEAD OR NECK

MANUAL TECHNIQUES

- MECHANICAL EXTERNAL FORCE:
  - SHAKING AND VIBRATIONS
  - PERCUSSION
  - MANUAL ASSISTED COUGH
- AIDS SHEARING FORCES AND EXPIRATORY FLOW FOR SECRETION CLEARANCE.
- NO STRONG EVIDENCE TO SUPPORT OR REJECT MANUAL TECHNIQUES COMPARED TO OTHER FORMS OF AIRWAY CLEARANCE
- CAN BE USEFUL IN INDIVIDUALS THAT UNABLE TO COLLABORATE IN TREATMENT SESSION



SHAKING AND VIBRATIONS

- NORMALLY COUPLED WITH POSITIONING
- HANDS ARE PLACED ON THE PATIENT'S CHEST
- SHAKING AND VIBRATIONS APPLIED DURING EXPIRATION PHASE OF BREATHING
- AIM TO CREATE OSCILLATION ON EXPIRATORY FLOW TO DISLODGE THICK MUCUS
- SHAKING : SLOW FREQUENCY
- VIBRATIONS: HIGH FREQUENCY



PERCUSSIONS

- DONE WITH CUPPED HANDS. REPEATED RHYTHMICAL FIRM PATS ON CHEST WALL
- COMPLETE WITH A LAYER OF FABRIC TO MINIMISE TRAUMA
- DURING TIDAL VOLUME
- FREQUENCY BETWEEN 4.6-8.6HZ
- EFFECTIVENESS IS DEPENDENT ON CHEST WALL COMPLIANCE



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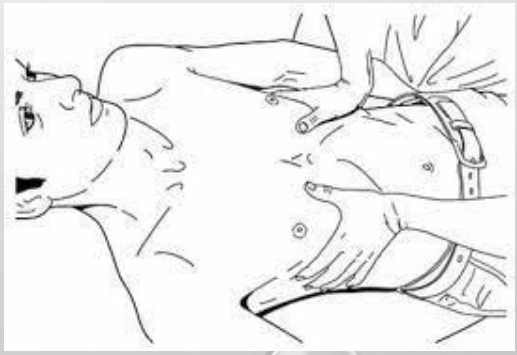
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MANUAL ASSISTED COUGH

- MOSTLY USED IN NEUROMUSCULAR CONDITIONS
- USES MANUAL TECHNIQUES TO AUGMENT A COUGH







MANUAL TECHNIQUES

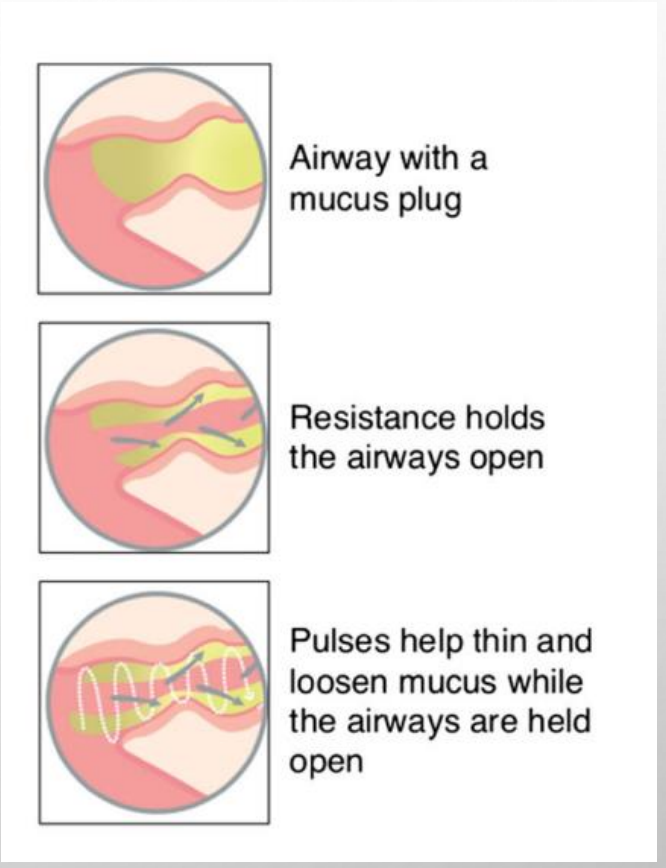
- INDICATIONS :
  - THICK SECRETIONS
  - SEDATED OR INTUBATED PATIENTS
  - PATIENTS WITH NEUROLOGICAL COMPROMISE
  - PATIENT UNABLE TO COOPERATE WITH SELF-ADMINISTERED TREATMENTS
  - INFANTS/TODDLERS

- CONTRAINDICATIONS:
  - RIB FRACTURES
  - OSTEOPOROSIS
  - RECENT FOOD INTAKE
  - TRAUMA

- CONSIDERATIONS :
  - SURGICAL RESTRICTIONS E.G STERNOTOMY
  - PAIN

PEP: POSITIVE EXPIRATORY PRESSURE

- STENTING OF AIRWAY
- INCREASE FUNCTIONAL RESIDUAL CAPACITY AND PREVENT EARLY COLLAPSE
- UTILISATION OF COLLATERAL VENTILATION
- MOST EFFECTIVE AT PRESSURE OF 10-20CMH2O
- STUDIES SHOWS THE NEED TO REACH TARGETED PRESSURE FOR 3-4 SECS PER BREATH TO BE EFFECTIVE
- CREATES BACK PRESSURE IN THE AIRWAYS DURING EXPIRATION
- ESPECIALLY USEFUL IN UNSTABLE AND REACTIVE AIRWAYS
- HOWEVER LIMITS EXPIRATORY FLOW BIAS



ADJUNCTS : PEP DEVICES AND OSCILLATING PEP

- PEP MASKS:
  - THE ORIGINAL PEP DEVICE
  - PATIENTS BREATHE OUT AGAINST A ONE-WAY VALVE WHICH PROVIDES PEP



ACT ADJUNCTS : OSCILLATING PEP







- DEVICES THAT PROVIDE PEP AND OSCILLATION
- SPLINTS OPEN AIRWAYS AND PROVIDES SHEARING FORCE TO AID IN SECRETION DISLODGEMENT
  - **ACAPELLA/AEROBIKA** : CAN BE USED IN MULTIPLE POSITIONS AND RESISTANCE CAN BE ALTERED, ALSO CAN ENTRAIN IN OTHER DEVICES
  - **FLUTTER** : GRAVITY DEPENDENT
  - **BUBBLE PEP**: MAINLY USED IN PAEDIATRICS AND POST SURGERY. TUBE PLACES IN WATER, PATIENTS THEN BLOW BUBBLES. THE AMOUNT OF WATER DICTATES THE RESISTANCE AND THEREFORE THE PEP. CAN BE LIMITED BY ASPIRATION RISKS IN PAEDIATRICS
- EVIDENCE SHOWS EFFECTIVENESS AT 11-15HZ

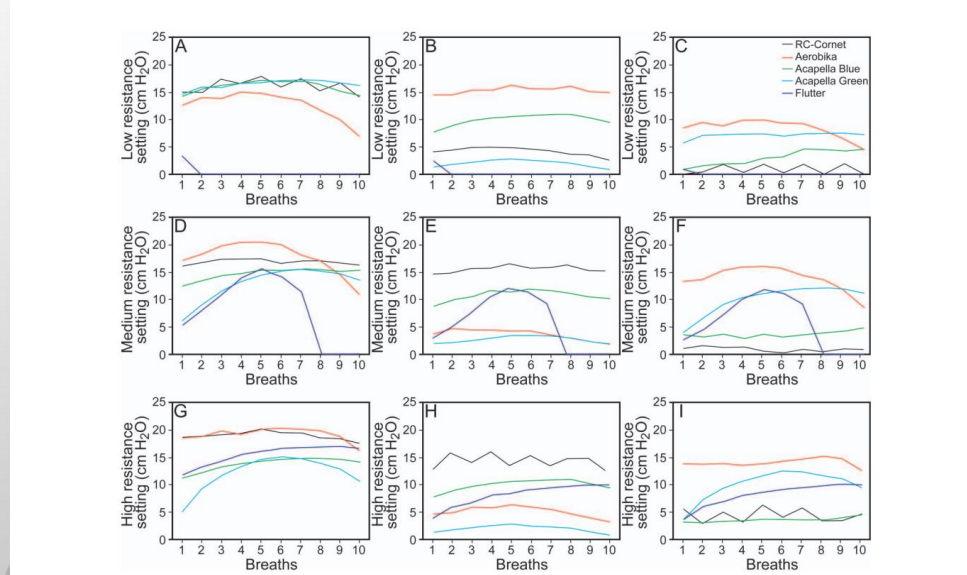


Fig. 4. Graphical display of mean values for peak pressure (A, D, G), positive expiratory pressure (B, E, H), and amplitude (C, F, I) for each device tested across the range of resistance settings.

PEP

- INDICATIONS:
  - THICK SECRETIONS
  - UNSTABLE AND REACTIVE AIRWAYS
  - HYPERINFLATION DUE TO AIRWAY OBSTRUCTIONS

- CONTRAINDICATIONS:
  - HAEMOPTYSIS
  - UNSAFE SWALLOW (FOR BUBBLE PEP)

IPPB

- IPPB : INTERMITTENT POSITIVE PRESSURE BREATHING.
  - POSITIVE PRESSURE IS PROVIDED BY THE MACHINE DURING INSPIRATION
  - EXPIRATION IS PASSIVE
  - INCREASES FRC, COLLATERAL VENTILATION AND INTERDEPENDENCE
  - DELIVERED VIA MOUTHPIECE OR FACE MASK
  - 3 MODIFIABLE FEATURES :
    - POSITIVE PRESSURE
    - SENSITIVITY
    - FLOW CONTROL
- CAN ADD IN ADJUNCTS SUCH AS PEEP VALVE AND OPEP.
- NEED TO ENTRAIN 0.9% SALINE NEBULISER SOLUTION TO AVOID DRY AIR
- STARTING TO BE PHASED OUT AS MACHINES ARE NO LONGER BEING REPAIRED OR SERVICED
- CAN BE REPLACED BY CLEARWAY 2 COUGH ASSIST



MANUAL INSUFFLATION : EXSUFFLATION / COUGH ASSIST

- USES POSITIVE AND NEGATIVE PRESSURE TO AUGMENT THORACIC EXPANSION AND COUGH EFFECTIVENESS
- DELIVERED VIA FACE MASK OR MOUTHPIECE
- TO ACHIEVE NET OUTWARD FLOW IN ORDER TO CLEAR SECRETIONS, THE EXPIRATORY FLOW MUST EXCEED THE INSPIRATORY FLOW
- MANY MODELS HAVE BOTH MANUAL AND AUTOMATIC MODES, SO TREATMENTS CAN BE PRESET FOR PATIENTS
- THERAPISTS CAN REGULATE:
  - PRESSURES OF INSUFFLATION AND EXSUFFLATION
  - LENGTH OF INSUFFLATION AND EXSUFFLATION
  - OSCILLATION
  - NUMBER OF INSUFFLATIONS PRIOR TO EXSUFFLATION







COUGH ASSIST AND IPPB

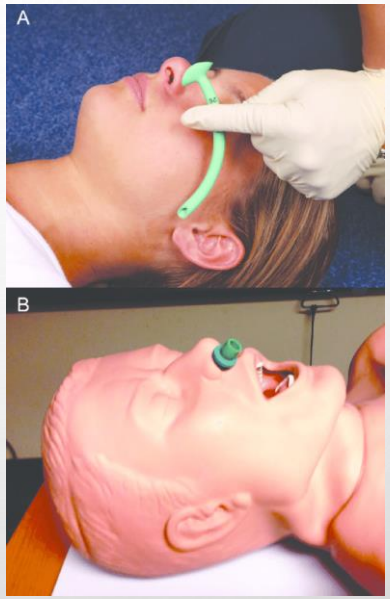
- INDICATIONS:
  - RETAINED SECRETIONS
  - STRUGGLES TO MOBILISE PERIPHERAL SECRETIONS
  - IMPAIRED COUGH (PAIN, NEUROLOGICAL COMPROMISE ETC)

- CONTRAINDICATIONS:
  - UNDRAINED PNEUMOTHORAX
  - HEAMOPTYSIS
  - BULLAE

- CONSIDERATIONS:
  - SURGICAL RESTRICTIONS IN PRESSURE
  - PAIN
  - CAN THEY MAKE A GOOD SEAL
  - VOMITING
  - RECENT THORACIC SURGERY : LOBECTOMY, PNEUMONECTOMY ANYTHING REQUIRING AN ANASTOMOSIS

INVASIVE PROCEDURES FOR ACT

- OROPHARYNGEAL AND NASOPHARYNGEAL SUCTION
- SHOULD BE A **FINAL RESORT**
- CLEAR SECRETIONS IN UPPER AND CENTRAL AIRWAYS
- CAN TRIGGER COUGH
- CAN CAUSE TRAUMA
- MAINLY USED IN EMERGENCY SITUATIONS OR PATIENTS WITH NEUROLOGICAL COMPROMISE
- OTHERWISE MUCH MORE EFFECTIVE IN CONJUNCTION WITH OTHER AIRWAY CLEARANCE TECHNIQUES
- CAN BE DONE VIA ARTIFICIAL AIRWAY OR "BLIND"
- THE KEY : CONFIDENCE AND SKILLS







INVASIVE PROCEDURES

- INDICATIONS:
  - UNABLE TO CLEAR SECRETIONS WITH OTHER ADJUNCTS OR TECHNIQUES
  - REDUCED LEVEL OF ALERTNESS AND UNABLE TO FOLLOW COMMANDS
  - UNABLE TO SPONTANEOUSLY OR ACTIVELY COUGH TO REMOVE SECRETIONS FROM CENTRAL AND UPPER AIRWAY
  - EMERGENCY TO PREVENT INTUBATION OR ESCALATION OF OXYGEN REQUIREMENT
- CONTRAINDICATIONS:
  - LOW PLATELETS
  - ACTIVE BLEEDING IN AIRWAY
- CONSIDERATIONS :
  - PATIENT CONSENT
  - PATIENT AGITATION LEVELS

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HONOURABLE MENTIONS









PHARMACOLOGICAL TREATMENTS

- BRONCHODILATORS
  - E.G SALBUTAMOL, ATROVENT, SYMBICORT



- MUCOLYTICS/MUCOREGULATORS/EXPECTORANTS/ MUCOKINETICS
  - HYPERTONIC SALINE
  - DORNASE ALFA
  - CARBOCISTEINE



HOW DO WE CHOOSE

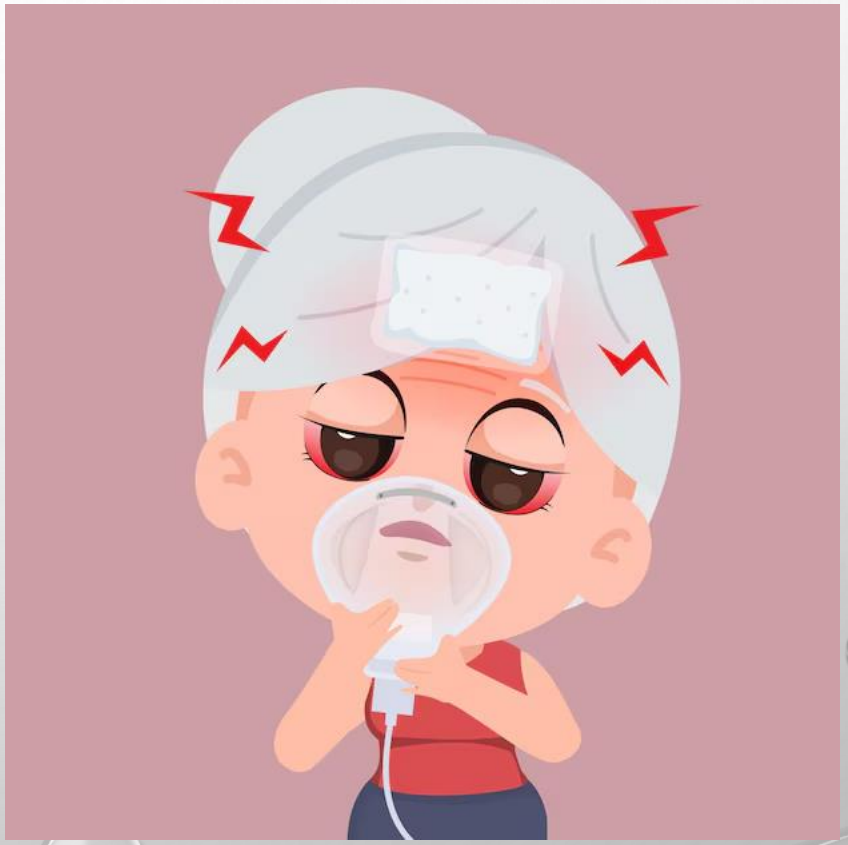
- PATIENT PREFERENCE
- EQUIPMENT AVAILABLE TO USE
- PATIENT ABILITY OR LEVEL OF CONSCIOUSNESS
- CONTRAINDICATIONS AND CONSIDERATIONS
- ACUTE / EMERGENCY

- EVIDENCE SHOWS THAT DOING AIRWAY CLEARANCE POORLY HAS THE SAME EFFECT AS NOT COMPLETING IT



CASE STUDY : MRS SILK

- MRS SILK 57F
- REFERRED BY THE NURSING STAFF
- ON NASAL CANNULA 4L BUT STRUGGLING
- DAY 1 POST STEP DOWN FROM ITU, ADMITTED 5/7 WITH PNEUMONIA
- NS REPORT REALLY THICK SECRETIONS, STRUGGLING TO CLEAR THEM



CASE STUDY : MRS SILK

YOU ARRIVE HER OBSERVATIONS ARE :

A: OWN, PATENT, AUDIBLE EXP CRACKLE

B: SV FIO2 35% VENTURE MASK, RR 30 SPO2 89%

AUSC: QUIET IN BASES, COARSE CRACKLES T.O + PALPABLE CRACKLES

C: HR 110 BP 120/80 T 36.5

D: AWAKE AND ALERT

E:PALE, LOOKS LIKE SHE IS WORKING HARD, ACCESSORY MUSCLES USE, PURSED LIP BREATHING

FELT THAT PHLEGM IS “STUCK” IN CHEST.

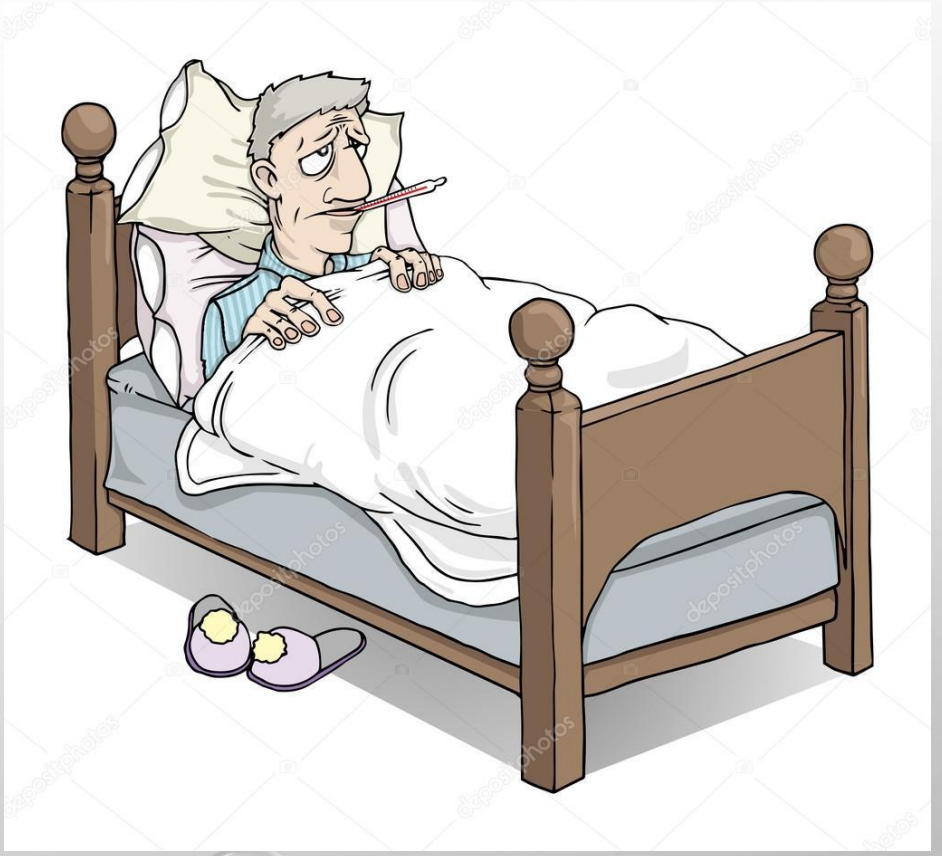






CASE STUDY : MR SAND

- MR SAND 62M
- HE IS DAY 2 POST CARDIAC SURGERY, EXTUBATED YESTERDAY AFTERNOON
- THE NURSES REPORT HE HAS BEEN SLOWLY DESATURATING OVER THE LAST 20 MINS AND NOW HE IS REQUIRING A HFNO FIO2 50%
- HE HAS NO HISTORY OF RESPIRATORY DISEASE AND IS NORMALLY VERY INDEPENDENT AND MOBILE



CASE STUDY : MR SAND

YOU EXAMINED HIM

A: OWN, PATENT

B: SV ON HFNO OF FIO2 50% RR 25

AUSC: VERY QUIET BASES. SOME AUDIBLE TRANSMITTED CRACKLES BILATERALLY

C: HR 80 BP 130/90 T 36.5

D: A BIT DROWSY BUT ROUSABLE AND ABLE TO FOLLOW INSTRUCTIONS

E: CENTRAL VENOUS LINE. DRAINS REMOVED THIS MORNING







CASE STUDY : MR SMITH

- MR SMITH 49M
- HE IS ADMITTED DUE TO CHEST INFECTION ON BACKGROUND OF MOTOR NEURON DISEASE
- THE NURSES REPORT ACUTE DESATURATION NOW SATURATING 87% ON 15L NON-REBREATHE BAG
- ESTABLISHED OF NON-INVASIVE VENTILATION WHICH HE NORMALLY USES NOCTURNALLY



CASE STUDY : MR SMITH

YOU EXAMINE HIM

A: OWN, PATENT

B: SV ON 15L VIA NRB RR 30

AUSC: NO AIR ENTRY HEARD IN THE RIGHT SIDE COMPLETELY

C: HR 105 BP 140/90 T 36.5

D: ALERT BUT UNABLE TO COMMUNICATE VERBALLY

E: HIGH WOB, UNABLE TO ACTIVELY COUGH







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QUESTIONS?



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